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3 Complex Analysis 2024

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Instructions:

- This question paper consists of **9 pages** (including this one).
- You have 120 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 60 points = 100% (63 points available).

Problem	1	2	3	4	5	6	7	Σ	Mark
Maximum Points	10	8	6	9	12	12	6	60	100%
Result									

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Problem 1

(10 points)

Find the set of all $z \in \mathbb{C}$ where the function $f(z) = |z - i|^2$ is differentiable.

Solution. We first show that f is not differentiable at any $z \neq i$.
Let $z = x + iy$ with $x = \operatorname{Re} z$, $y = \operatorname{Im} z$, and let $f = u + iv$ with $u = \operatorname{Re} f$, $v = \operatorname{Im} f$. We have

$$f(z) = |z - i|^2 = |x + i(y - 1)|^2 = x^2 + (y - 1)^2,$$

and therefore

$$u(x, y) = x^2 + (y - 1)^2, \quad v(x, y) = 0.$$

The derivatives are

$$u_x = 2x, \quad u_y = 2(y - 1), \quad v_x = 0, \quad v_y = 0.$$

The Cauchy-Riemann equations $u_x = v_y$ and $u_y = -v_x$ take the form

$$2x = 0, \quad 2(y - 1) = 0.$$

The only solution is $(x, y) = (0, 1)$. It follows that f is not differentiable at any $z \neq i$.

Now let us consider $z = i$. We have

$$\lim_{h \to 0} \frac{f(i + h) - f(i)}{h} = \lim_{h \to 0} \frac{|h|^2 - 0^2}{h} = \lim_{h \to 0} \frac{|h|^2}{h}.$$

We claim that $\lim_{h \to 0} \frac{|h|^2}{h} = 0$. Indeed,

$$\lim_{h \to 0} \left| \frac{|h|^2}{h} - 0 \right| = \lim_{h \to 0} |h| = 0.$$

Thus, f is differentiable at $z = i$ (with $f'(i) = 0$).

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Problem 2

(8 points)

Calculate the following integrals.

(a) $\displaystyle\int_{\gamma_1} \frac{1}{(z+i)^3}\, dz$, where $\gamma_1(t) = 2e^{it}$, $t \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

(b) $\displaystyle\int_{\gamma_2} \frac{1}{(z+i)^3}\, dz$, where $\gamma_2(t) = 2e^{it}$, $t \in [-\pi, \pi]$.

Solution.

(a) The function $f(z) = \frac{1}{(z+i)^3}$ has an antiderivative $F(z) = -\frac{1}{2(z+i)^2}$ in the open set $\mathbb{C} \setminus \{-i\}$. The path γ_1 is a path in $\mathbb{C} \setminus \{-i\}$, it is a semicircle of radius 2 about the origin with endpoints $\gamma_1\left(-\frac{\pi}{2}\right) = -2i$ and $\gamma_1\left(\frac{\pi}{2}\right) = 2i$. By the Fundamental Theorem of Calculus,

$$\int_{\gamma_1} \frac{1}{(z+i)^3}\, dz = F(2i) - F(-2i) = -\frac{1}{2(2i+i)^2} + \frac{1}{2(-2i+i)^2} = -\frac{1}{2(3i)^2} + \frac{1}{2(-i)^2}$$
$$= \frac{1}{2\cdot 9} - \frac{1}{2} = -\frac{1}{2}\left(1 - \frac{1}{9}\right) = -\frac{4}{9}.$$

(b) The function $f(z) = \frac{1}{(z+i)^3}$ has one singularity $z_0 = -i$ inside γ_2 which is a pole of order 3. By the Residue Theorem

$$\int_{\gamma_2} \frac{1}{(z+i)^3}\, dz = 2\pi i \operatorname{Res}(f, -i) = \frac{2\pi i}{2!}\left((z+i)^3 f(z)\right)''\Big|_{z=-i} = \pi i\,(1)''\Big|_{z=-i} = 0.$$

(Alternatively, one sees immediately that $\operatorname{Res}(f, -i) = 0$ because the Laurent coefficient a_{-1} of the function $f(z) = \frac{1}{(z+i)^3}$ is zero.)

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Problem 3

(6 points)

Does there exist a holomorphic function $f : \mathbb{C} \to \mathbb{C}$ with

$$f\left(\frac{1}{n}\right) = \frac{1}{n} \quad \text{for all} \quad n \in \mathbb{N}, \qquad f(-1) = 1?$$

Solution.
The set $\left\{\frac{1}{n} : n \in \mathbb{N}\right\}$ has a limit point $0 \in \mathbb{C}$.

If f is holomorphic on $\mathbb{C}$ and $f\left(\frac{1}{n}\right) = \frac{1}{n}$, $n \in \mathbb{N}$, then by the Coincidence Principle f coincides in $\mathbb{C}$ with the function $f(z) = z$.

But then we would have

$$f(-1) = -1 \neq 1.$$

We conclude that there is no such function f.

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Problem 4

(9 points)

Use methods of complex analysis to calculate the integral $\displaystyle\int_0^{2\pi} \frac{1}{5-3\cos\theta}\, d\theta$.

Solution. Using the formula $\cos\theta = \frac{1}{2}(e^{i\theta}+e^{-i\theta})$, we write

$$\begin{aligned} I &= \int_0^{2\pi} \frac{1}{5-3\cos\theta}\, d\theta = \int_0^{2\pi} \frac{1}{5-\frac{3}{2}(e^{i\theta}+e^{-i\theta})}\, d\theta = \int_0^{2\pi} \frac{2}{10-3(e^{i\theta}+e^{-i\theta})}\, d\theta \\ &= \int_0^{2\pi} \frac{2e^{i\theta}}{-3e^{i2\theta}+10e^{i\theta}-3}\, d\theta. \end{aligned}$$

If θ runs from 0 to 2π, the term $z = e^{i\theta}$ traces the unit circle $\gamma(\theta) = e^{i\theta}$. We have $\gamma'(\theta) = ie^{i\theta}$. Therefore,

$$I = \int_0^{2\pi} \frac{2\gamma'(\theta)}{-3(\gamma(\theta))^2+10\gamma(\theta)-3}\, d\theta = \frac{1}{i}\int_\gamma \frac{2}{-3z^2+10z-3}\, dz.$$

The zeros of the polynomial $-3z^2+10z-3$ are

$$z_{1,2} = \frac{-10 \pm \sqrt{100-36}}{-6} = \frac{10\pm 8}{6}, \qquad z_1 = \frac{1}{3}, \quad z_2 = 3.$$

The function $f(z) = \frac{2}{-3z^2+10z-3}$ has two simple poles at $z_1 = \frac{1}{3}$ and $z_2 = 3$. Only $z_1 = \frac{1}{3}$ lies inside γ. We obtain

$$\begin{aligned} I &= \frac{1}{i} 2\pi i \operatorname{Res}\left(\frac{2}{-3z^2+10z-3}, \frac{1}{3}\right) = 2\pi \left.\frac{2}{(-3z^2+10z-3)'}\right|_{z=\frac{1}{3}} \\ &= \left.\frac{4\pi}{-6z+10}\right|_{z=\frac{1}{3}} = \frac{4\pi}{-2+10} = \frac{\pi}{2}. \end{aligned}$$

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Problem 5

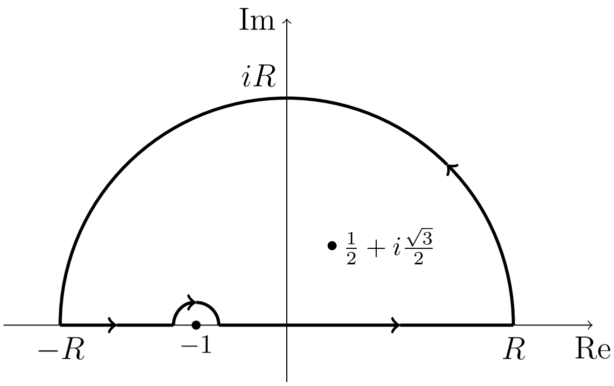
(12 points)

Use methods of complex analysis to calculate P.V. $\int_{-\infty}^{\infty} \frac{1}{x^3+1}\,dx$.

Solution.
The zeros of the polynomial z^3+1 are the 3th roots of $-1 = e^{i\pi}$, namely,

$$\begin{aligned} z_0 &= e^{i\frac{\pi}{3}} = \cos\frac{\pi}{3} + i\sin\frac{\pi}{3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}, \\ z_1 &= e^{i\frac{3\pi}{3}} = -1, \\ z_2 &= \overline{z_0} = \frac{1}{2} - i\frac{\sqrt{3}}{2}. \end{aligned}$$

They are all simple poles.
Consider the path γ indicated below with R large enough:



Let $\gamma_R(t) = Re^{it}$, $t \in [0,\pi]$ be the large semicircle with radius R,
and let $\gamma_\varepsilon(t) = -1 + \varepsilon e^{i(\pi - t)}$, $t \in [0,\pi]$, be the small semicircle with radius $\varepsilon > 0$ and center -1, negatively oriented.
We have

$$\begin{aligned} \int_\gamma \frac{1}{z^3+1}\,dz &= 2\pi i \operatorname{Res}\left(\frac{1}{z^3+1}, \frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 2\pi i \left.\frac{1}{3z^2}\right|_{z=\frac{1}{2}+i\frac{\sqrt{3}}{2}} \\ &= \frac{2\pi i}{3\left(\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)^2} = \frac{2\pi i}{3\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)} = \frac{2\pi i}{3}\cdot\frac{-\frac{1}{2}-i\frac{\sqrt{3}}{2}}{\frac{1}{4}+\frac{3}{4}} = -\frac{\pi i}{3} + \frac{\pi\sqrt{3}}{3}. \end{aligned}$$

On the other hand,

$$\int_\gamma \frac{1}{z^3+1}\,dz = \left(\int_{\gamma_R} + \int_{[-R,-1-\varepsilon]} + \int_{\gamma_\varepsilon} + \int_{[-1+\varepsilon,R]}\right) \frac{1}{z^3+1}\,dz.$$

If $|z| = R$, then $|z^3+1| \geq |z|^3 - 1 = R^3 - 1$. We estimate

$$\left|\int_{\gamma_R} \frac{1}{z^3+1}\,dz\right| \leq \frac{1}{R^3-1}\pi R \to 0, \quad R \to \infty.$$

Furthermore,

$$\lim_{\varepsilon\to+0} \int_{\gamma_\varepsilon} \frac{1}{z^3+1}\,dz = -\pi i \operatorname{Res}\left(\frac{1}{z^3+1}, -1\right) = -\pi i \left.\frac{1}{3z^2}\right|_{z=-1} = -\frac{\pi i}{3}.$$

It follows that

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{1}{x^3+1}\, dx = \lim_{\substack{R\to\infty \\ \varepsilon\to+0}} \left(\int_{[-R,-1-\varepsilon]} + \int_{[-1+\varepsilon,R]} \right) \frac{1}{z^3+1}\, dz$$

$$= -\frac{\pi i}{3} + \frac{\pi\sqrt{3}}{3} - \left(-\frac{\pi i}{3}\right) = \frac{\pi\sqrt{3}}{3}.$$

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Problem 6

(12 points)

(a) Find the radius of convergence of the power series $\sum_{n=0}^{\infty} \frac{(n!)^3}{(3n)!} z^{3n}$.

(b) Determine the Laurent series of the function $f(z) = \frac{1}{9-4z^2}$ in the region $\left\{z \in \mathbb{C} : |z| > \frac{3}{2}\right\}$. Give an explicit formula for the coefficients a_n, $n \in \mathbb{Z}$, of the Laurent series.

Solution.

(a) We are going to use the Ratio Test for the real series $\sum_{n=0}^{\infty} \frac{(n!)^3}{(3n)!}|z|^{3n}$. We have

$$\begin{aligned} L &= \lim_{n\to\infty} \frac{((n+1)!)^3|z|^{3n+3}}{(3n+3)!} \cdot \frac{(3n)!}{(n!)^3|z|^{3n}} = |z|^3 \lim_{n\to\infty} \frac{(n+1)^3}{(3n+3)(3n+2)(3n+1)} \\ &= \frac{|z|^3}{3^3} < 1 \iff |z|^3 < 3^3 \iff |z| < 3. \end{aligned}$$

The radius of convergence of the series $\sum_{n=0}^{\infty} \frac{(n!)^3}{(3n)!} z^{3n}$ is $R = 3$.

(b) We have

$$\begin{aligned} f(z) = \frac{1}{9-4z^2} &= \frac{1}{-4z^2\left(1 - \frac{9}{4z^2}\right)} = \frac{-1}{4z^2} \cdot \sum_{n=0}^{\infty} \left(\frac{9}{4z^2}\right)^n \\ &= -\sum_{n=0}^{\infty} \frac{9^n}{(4z^2)^{n+1}} = -\sum_{n=0}^{\infty} 9^n 4^{-(n+1)} z^{-2(n+1)} = -\sum_{n=1}^{\infty} 9^{n-1} 4^{-n} z^{-2n}. \end{aligned}$$

The coefficients of the Laurent series are, for $m \in \mathbb{Z}$,

$$a_m = \begin{cases} 0, & m \geq 0 \text{ or } m \text{ odd}, \\ -9^{n-1}4^{-n}, & m = -2n \text{ with } n \in \mathbb{N}. \end{cases}$$

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Problem 7

(6 points)

How many zeros does the polynomial $p(z) = z^{10} + z^5 + z^2 + 5$ have in the closed disc $\{z \in \mathbb{C} : |z| \leq 1\}$?

Solution.

We consider $f(z) = 5$ and $g(z) = z^{10} + z^5 + z^2$. On the circle $|z| = 1$ we have

$$|f(z)| = 5, \quad |g(z)| \leq |z|^{10} + |z|^5 + |z|^2 \leq 3.$$

Thus,

$$|g(z)| < |f(z)|, \quad |z| = 1.$$

By Rouche's Theorem, f and $p = f+g$ have the same number of zeros in $\{z \in \mathbb{C} : |z| < 1\}$. Clearly, $f(z) = 5$ has no zeros in $\{z \in \mathbb{C} : |z| < 1\}$. We conclude that p has no zeros in $\{z \in \mathbb{C} : |z| < 1\}$.

Since $|g(z)| < |f(z)|$ on the circle $|z| = 1$, we see that $p = f + g$ cannot have zeros with $|z| = 1$.

We conclude that p has no zeros in the closed disc $\{z \in \mathbb{C} : |z| \leq 1\}$.